Fusion of Local Convolutional Features and Global Attention for Automated Leukemia Cell Detection

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I. ABSTRACT

Microscopic blood smear images have been proven to be promising in the early diagnosis of leukemia with the use of automated analysis [1,2,3]. Nonetheless, a great number of current systems that are based on deep learning ecstatically broadcast their effectiveness in restricted evaluation contexts and are unstable whenever they are utilized in alternative data divisions or image circumstances [11,12]. This work is a hybrid convolutional neural network (CNN)-Transformer architecture which was trained to perform binary classification between healthy cells and Acute Lymphoblastic Leukemia (ALL). The proposed system is focused on the training stability, cross-validation folds across stratified cross-validation, and interpretability as opposed to the peak single-run accuracy. Local feature extraction is performed with the help of a ResNet50 backbone [15] , after which features are converted into tokens and processed on the feature level with the help of transformers to refine the results [18,19] . Self-supervised feature-level masked image modeling (MIM) pretraining stage is further investigated to further reinforce representations when limited labeled data are used [18] . Optuna is used to select models and optimize hyperparameters through fold-wise consistency and a held-out test set is used to evaluate the final performance [22,23]. Moreover, a combination of Grad-CAM and transformer attention visualizations is used to analyze model decisions to achieve complementary interpretability [21]. The experimental outcomes prove that the specified methodology provides a stable output over folds as well as provides valuable information about the model behavior, which is why it can be applied in the context of leukemia screening in real-world settings [1,3].

Keywords: Leukemia Detection · CNN · Transformer · Medical Image Analysis

II. INTRODUCTION

Leukemia is a cancerous complication of the blood-forming tissues demanding rapid and precise diagnosis to enhance the outcome of the patients [1,2]. The microscopic analysis of blood smear pictures is an important element of leukemia screening, whereas manual analysis is not only time-consuming but also has a tendency to be inter-observer variability. To address these issues, there is a growing interest in automated image-based solutions that would be able to help clinicians by offering quick and reliable evaluations [4], [5].The latest developments of deep learning have leaped the automated leukemia detection system performance [1,4,9,13]. The convolutional neural network (CNN) in particular has exhibited a high ability in its discernment to visual learn discriminatory features on cellular images [15,16,17]. Consequently, a number of computer-aided diagnosis pipelines in the field of hematology are now based on CNN-based models [5,6,7,8,9,10]. Even with these developments, it is difficult to convert high validation accuracy to realistic work world performance [6,11].The first of the challenges is in the variability in nature of blood smear images. Varied staining conditions, lighting conditions and cell morphology presuppose considerable visual variability even within the same category of diagnosis [3,6]. Also, publicly available datasets of leukemia are commonly quite small and are unbalanced in terms of the classes, which results in overfitting models and unstable training [6,11]. As a result, the systems that work well with a given data split are not always able to generalize with unobserved samples.Most recently, medical imaging architecture has turned to transformer-based architectures to overcome the drawbacks

of purely convolutional architectures [18,19]. Transformers use long-range dependencies to provide a way to use global context in image representations [19]. Nevertheless, methods based on transformer usually involve a close architectural design and training plans particularly in cases where relatively small datasets in the medical field are involved. Simplistic application of transformers at either the image or patch level may apply convergence and be hyper-sensitive to hyperparameter decisions [18,19].This paper presents our hybrid CNNTransformer architecture which is more robust, more stable, and more interpretable in detecting leukemia. Instead of making the system optimize to achieve the best accuracy in one run, the suggested system is to learn the representations that show consistent results in stratified cross-validation folds [22,23]. The feature-level tokenization is used to maintain the semantic structure, but allow the contextual modeling using transformers [18,19]. Optuna [26] is used to perform hyperparameter optimization as a held-out test set is held back to perform final evaluation [22,23]. In order to aid clinical relevance, model predictions are considered with supplementary explainability methods [21].The rest of this paper is structured in the following way. Section II provides a literature review of leukemia detection using deep learning. Section III talks about the limitations that are commonly encountered in current systems. Section IV will provide the proposed methodology that will cover dataset management, model design, training plan, and explainability. Results and analysis are then given based on the experimental results and finally discussed and conclusions made.

III. Literature review

In recent years, much attention has been paid to automated detection of leukemia in microscopic blood smear images with the development of deep learning and the existence of publicly available datasets [4,5,6,7]. The initial research mostly consisted of hand-crafted characteristics and the old-fashioned machine learning classifier [7,16,19]. Although these methods proved to be initially viable, they could only perform well due to the challenge of creating resistant features that are able to generalize in the case of staining, illumination, and cell morphology variation [3,6].As convolutional neural networks (CNNs) excelled at analysing medical images, ensuing research switched to an end-to-end architecture that uses CNNs to classify leukemia [7,8,9,10]. Widespread designs included VGG, ResNet and different variations of the Inception, typically pretrained on ImageNet and then trained on leukemia data [15,16,17]. These models were found to be more accurate, as they make use of hierarchical visual representations as they learn directly off data [12,13]. The CNN-based systems were found to be effective in a number of cases in curated validation sets that support the effectiveness of the systems in cellular image analysis [7,8,9,10]. More recent literature has delved into the architectural differences and refinements to continue the classification findings through better classification [2,21,23]. There are studies that use lightweight CNNs to minimize the number of computations and research that explored more comprehensive or richer networks to represent finer details of the nucleus [9,10]. Preprocessing and data augmentation methods, such as color normalization and stain correction, have also been tried to resolve visual variability of blood smear images [14]. Furthermore, ensemble strategies have been added to a list of works aimed at enhancing the performance through a pool of predictions made by multiple models [2,4].Transformer-based models have also started to be used on leukemia detection and related histopathology tasks driven by their long-range dependency modeling capabilities [18,19]. In certain works, transformers are used with image patches, whereas in others, they are used with convolutional layers in hybrid implementations [18,19]. These methods attempt to supplement local feature detection using global contextual reasoning [19]. Nevertheless, transformer models tend to be harder to train and generally need to be fine-tuned not to overfit, especially when the size of the datasets is small [18,19].The literature in the field of evaluation practices differs significantly. Some of the studies use a cross-validation, whereas others use a single train/test split which makes it hard to directly compare the two [22,23]. Accuracy, precision, recall and F1-score are often reported measures, but not all studies give detailed information on measures of variability between runs or splits [22]. Consequently, the increases in reported performance are not always easy to explain in the context of robustness and generalization [23,24].Several works have also dealt with explainability, most frequently by using gradient-based visualization techniques like Grad-CAM [21]. The techniques can give qualitative information on the areas of interest that affect model predictions and are commonly tabled in support of model reliability [21]. Explainability is however considered as a post hoc analysis and not as part and parcel of the system design [21].In general, the current literature shows that deep learning-based solutions have the potential to detect leukemia and investigate a diverse set of architecture and training schemes [12,13,18]. These researches offer a platform on which more solid and clinically consistent systems can be structured [1,2].

Although some of the studies conducted in the past have shown encouraging outcomes, several limitations are common when such systems are looked into through the deployment and generalization prism [5,6,7].The most recent studies of automated leukemia detection with microscopic images of blood smears have demonstrated that deep learning-based methods can perform well in classification when it is controlled [7,8,9,10]. Upon close examination of these systems however, it is possible to find a number of structural constraints to these systems, which do not only undermine their integrity and dependability in real world applications [6,11].A feature that is shared by most current systems is the high degree of dependence on CNN-only infrastructure [15,16,17]. Even though convolutional models are useful at representing local texture and shape detail, they are not, in principle, able to represent long-range spatial relations in an image [18,19]. Most methods implicitly presuppose that localized nuclear features can be relied on to make a diagnosis [3,6]. Practically, diagnostically relevant cues are frequently interactions between several spatial regions, which CNN-based models are not intermittently able to detect [18,19]. In recent work, transformer-based components have been proposed, which are typically used on a pixel or patch scale, resulting in unstable dynamics of training and being sensitive to hyperparameter decisions [18,19].Another issue that arises is evaluation methodology. A number of the available studies provide their results based

on one train- validation split or on a small-scale cross-validation not being strictly stratified [22,23]. Since the leukemia datasets have a class imbalance and heterogeneity on the subject level, these types of evaluation can result in a positive prediction of the performance [11,22]. Little to no discussion on variation between folds or sensitivity to data partitioning is generally given and it is hard to determine whether or not any improvements being reported are an actual generalization or a good split selection [22,23].Another area where most of the previous systems have not been well covered is the variability of datasets. Variations in staining procedures, lighting settings, and cell shape are not normally taken into consideration when making building or training choices [3,6,14]. Consequently, there are models that can be extremely accurate on carefully curated validation sets but perform badly when presented with visually different samples [6,11]. Moreover, most of the current models are only based on supervised learning, limiting the quality of representation when there is limited and unevenly distributed labeled data [18].Another place where the existing systems are inadequate is explainability. Although saliency visualizations are a part of some studies, they are usually computed based on a single model component and are usually presented in the form of qualitative examples and not as an integrated analytical tool [21]. This restricts their practical use in the clinical setting, where local evidence and large scale contextual inferences are significant to interpret model predictions [1,3]. The explanation mechanisms are often considered optional features of the system, which should not be regarded as its central aspects [21].The limitations are solved in the proposed system by a set of focused design decisions. An architecture based on hybrid CNN-Transformer is used to integrate the strength of local feature production with the contextual modeling on a global scale, making it more stable than a transformer-only architecture [18,19]. The level of tokenization is applied on features, rather than on pixel-level patches, to make the noise less sensitive and convergence more favorable [18]. The focus of models selection and hyperparameter optimization is on fold-wise consistency based on stratified cross-validation as opposed to peak performance in a single run [22,23]. Another feature-level pretraining phase that is self-supervised is also investigated to enhance the representations with low benchmarks of labelled data [18]. Lastly, it improves interpretability through integrating CNN-based Grad-CAM display with transformer attention analysis, which offers complementary insight into the decisions of a model instead of using only one explanation mechanism [21].

IV. Methodology

A. Problem Definition & Learning Objectives :

The proposed study aims at designing a deep learning architecture to detect leukemia [5,6,7]. Although the general task is conceptually easy, training stability is a great challenge [6,11]. The problem is defined as a binary classification problem to differentiate healthy cells (HEM) and the Acute Lymphoblastic Leukemia (ALL) [1,2]. The main issues are that there were subtle inter-class differences, differences in staining procedures, and strong inter-individual differences among the data [3,6]. To evaluate the efficacy of the model and also to prove the clinical utility, area under the receiver operating characteristic curve (AUC-ROC) is taken as the primary success measure [22,23]. This measure is a single numerical measure of the capacity of the model to differentiate between classes and is popular in clinical settings [22]. The goal of the learning process is not only to maximize the classification accuracy [23,24]. First trials suggested that some models worked effectively on particular validation splits but had poor generalization [6,11]. Even slight differences in how the initial samples were set up would frequently lead to large divergences in the validity of single-split assessments, casting doubt on the consistency of single-split assessments [22,23]. As a result, attention shifted to the creation of stable and interpretable representations with good extrapolation across several data splits [18,21]. Interpretability was also emphasized because there is a major necessity of effective leukemia screening [1,3].

B. Description of Data set and Split Strategy :

The C-NMC Leukemia data were used in experiments, comprising of labeled images of healthy and leukemic blood cells [11]. This data is class imbalanced, and contains a lot of visual variation due to the differences in the staining, the lighting and the shape of the cells [3,6]. All these things complicate training strategies and lead to overfitting [6,11].In order to minimize the effects of splits bias, we applied a stratified three-fold cross-validation approach when developing the model [22,23]. We ensured that there was consistency in the class proportions within the same fold to obtain balanced training and validation [22]. The final assessment involved a distinct set of tests that were not utilized in any way in model selection or hyperparameter tuning [23]. In our initial experiments we observed that non-stratified splits may swell run-to-run performance disparities [22,23]. This contributed to the significance of a controlled splitting strategy [22]. This approach prevents the selection of the results and reflects more closely the performance in practice [23].

C. General Architecture Review :

The last model is an implementation of a hybrid CNN-Transformer model to balance between local feature extraction and global context [18,19]. The CNN-only models were initially tested and did a good job in capturing local patterns but failed in dealing with distant relationships [18,19]. We have, as well, experimented with transformer-only methods, which had training issues, and tended to give unstable results, or tend to overfit easily [18,19].The hybrid design is a compromising solution [18,19]. A convolutional backbone obtains high-level feature representations, and a transformer encoder enhances them by feature tokens that are projected [18,19]. Full variants of these experiments were more sensitive to learning rate and initialisation than anticipated and a number of the experiment runs did not converge cleanly [18,19]. Instead, we had the architecture as modular than tightly coupled [19]. This separation enhanced the stability of training and has facilitated the debug of the studies conducted in experimentations [19].

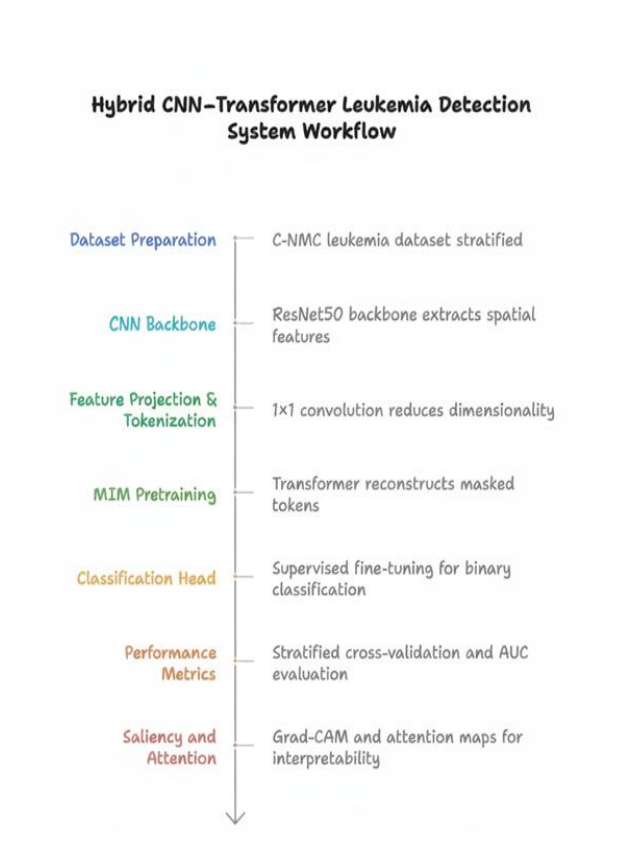


Fig 1. A hybrid CNN-Transformer that is trained on ResNet-50 and MIM at the start of leukemia detection on C-NMC data, measured with stratified cross-validation, AUC, Grad-CAM, and attention maps.

D. CNN Backbone for Local Feature Extraction :

The convolutional backbone used to extract features was identified as a ResNet50 that was trained on ImageNet [15]. This dropped the last fully connected layer in order to store the intermediate feature maps [15]. Maintaining spatial information at this point was significant in the context of acquiring cell morphology that would be important in detecting leukemia [3,6].Some of the other backbones that were explored in early experiments are ResNet18 and EfficientNetB0 [15,23]. ResNet18 would converge quickly but frequently would not provide finer details of the nucleus, particularly when the sample had subtle abnormalities [6,9]. EfficientNetB0 has occasionally made better validation accuracy per-run [23]. Nevertheless, it showed a significant cross-validation fold variation [22,23]. Hyperparameter tuning with optuna revealed that EfficientNet models were very sensitive to the choice of learning rate and regularization [22,23]. There are a few instances where training behavior is unstable with the change of small parameters [22,23]. To have some form of sanity check to test EfficientNet stability claims, we suggest taking three random seeds and training and evaluating the performance variance between these three seeds [22,23]. Readers can repeat this basic diagnostic test to confirm the impact of hyperparameter fragility [22]. By comparison, however, ResNet50 had more stable folds, thus its choice in spite of the increased computational cost [15].

E. Feature Projection & Tokenization :

The CNN backbone generates feature maps which are of high dimensions and cannot be directly inputted into an encoder of a transformer [18,19]. In order to overcome this, a learnable 1x1 convolution was used to map the feature maps and project them to a reduced dimensional embedding space [15,18]. This estimate minimised the computational load whilst holding the majority of the semantic content acquired by the convolutional layers [15]. The feature maps projected were then reformed into a series of tokens [18,19]. The tokens are spatial units of the original image [18]. The feature level tokenization was preferred to pixel-level tokenization [18,19]. Initial experiments with raw image patches had slower convergence and were also more sensitive to noise [18,19]. Training was found to be unstable in certain of the runs unless conservative learning rates were employed [18,19]. Keeping tokenization at the level of feature-map offered a more stable and structured input to transformer based modeling [18,19].

F. Transformer Encoder for Global Context Modeling :

The transformer encoder was built in as a contextual refinement module which acts on the order of feature tokens [18,19]. Multi-head self-attention helps the model to capture connections between spatially remote areas that can collectively affect the diagnostic results [19]. This was quite lacking in CNN-only baselines [18,19].Transformer layers and attention head count were studied through the application of hyperparameter optimization based on Optuna [19]. The deeper configurations were found to minimize the training loss though there was indication of overfitting in the cross-validation [22, 23]. Explicit positional embeddings were also tested. Nevertheless, they were not always able to enhance performance and increased validation variance between folds [22, 23]. Some of the trials were more stable when positional embeddings were removed [19]. Consequently, positional information was stored implicitly in the form of the token position and the last configuration preferred moderate depth and attention width [18,19].

G. Self-Supervised Pretraining via Feature-Level Masked Image Modeling (MIM) :

In order to enhance representation learning with fewer labeled data, self-supervised pretraining with feature-level Masked Image Modeling (MIM) was investigated [18]. Here, a CNN backbone had been frozen and a sub-set of projected feature tokens had been masked randomly [18,19]. Transformer encoder was trained to recover the masked features with the loss of mean squared error [18].The masking ratio was a sensitive parameter [18,19]. Increased masking rates resulted in convergence problems and the reduced rates resulted in minimal value [18]. Some of the initial runs could not stabilize with violent masking [18,19]. An empirical tuning technique was used to settle on a moderate masking strategy [18,19]. This has enabled the model to acquire contextual relations without collapsing the training [18,19]. The MIM step was considered as an experiment addition and it was maintained when it enhanced consistency in the process of supervised fine-tuning [18].

H. *Supervised Fine-Tuning for Classification :*

The entire model was then fine tuned after self-supervised pretraining to supervised classification [18]. On the transformer output, a lightweight classification head was connected which was comprised of two fully connected layers with dropout to ensure regularization [18,21]. The reason this design was adopted was to restrict the model capacity and minimize the chances of over fitting, due to the size of the dataset [6,11].The conservative learning rates were applied during the fine-tuning to keep the pretrained representations [18]. The rate of learning was more aggressive, and it dominated the faster initial performance but caused erratic validation behavior [22,23]. During various initial runs the training accuracy improved with each training step, and validation accuracy varied with the epoch [22]. To counter this, the learning rates and weight decay values were maintained within the small ranges because the focus was made on stability as opposed to fast convergence [18].

I. *Training Strategy & Cross-Validation Protocol :*

A stratified three-fold cross-validation protocol was used to conduct model training, so that there were equal class distributions between splits [22,23]. The folds were trained separately and reported the performance as the mean fold performance [22]. This was done to minimize the variance that would be caused by specific splits and to further gauge the behavior of generalization [22,23].The hyperparameters that were optimized with the help of Optuna in the cross-validation system were learning rate, batch size, and regularization strength [22,23]. No data on the set ofheld-out was utilized in this [23]. The early experiments found that increasing the batch sizes resulted in faster training but less stable fold-wise validation [22,23]. Consequently, the smaller batch sizes were preferred even though the training took greater periods [22]. The choice of models was encouraged to focus on uniform cross-validation on performance instead of the highest accuracy on one fold [22,23].

J. *Inference Pipeline :*

In inference, a single forward pass was used, directly on untested test images, on the trained model [22,23]. None of the test-time augmentation or model ensembling took place [22]. This choice was chosen because it would be more realistic in the real world of deployment, where latency and computational constraints can be very significant [12,13].Test-time augmentation in preliminary experiments yielded slight increase in performance on certain samples [24], but produced variation between runs [22,24]. In some instances, it varied according to the strategy of augmentation with predictions of borderline samples. In order to mitigate such inconsistency, a less complex inference pipeline was used and reproducibility and interpretability were more valued than gain a marginal accuracy [22,23].

K. *Explainability Module (Grad-CAM + Attention Fusion):*

The interpretability of the models was solved by integrating Grad-CAM visualizations using the CNN backbone with attention-based maps using the transformer encoder [21]. Grad-CAM can point out regions in an image that have a strong effect on the convolutional features activation [21], and the transformer attention allows one to see how spatial areas interrelate on a global scale [19].The two types of visualization were compared and the results were obtained to have a complementary view of model behavior [21]. In other instances, the localized nuclear regions were the focus of Grad-CAM and attention maps were more general [1,3]. But there were also inconsistencies between the two especially with the ambiguous samples [21]. These differences were not enforced; instead they represented a sign of uncertainty which supports the rationale of cautionary interpretation in clinical situations [1,3].

V. RESULT

A. *Quantitative Performance Analysis :*

The hybrid CNNTransformer model was tested on stratified three-fold cross-validation on the C-NMC leukemia data in the Methodology section [11,22,23]. The receiver operating characteristic (ROC) curve and area under the receiver operating characteristic (ROC–AUC) were the main techniques used to evaluate model performance because these are threshold-free measures of classification performance and are most applicable when medical diagnostic tasks are to be addressed [22,23].

The summary of the fold-wise ROC-AUC values on the validation set is presented below:

I) **Fold 1:** 0.719

II) **Fold 2:** 0.815

III) **Fold 3:** 0.730

The overall results are consistent with an average ROC -AUC of 0.755 and a standard deviation of 0.043, which indicates that there is homogeneity in the discriminative performance of the folds [22,23]. In both scenarios, the ROC curves are very far above the random baseline (ROC -AUC = 0.50) and this ensures that the model is indeed learning meaningful morphological differences between healthy (HEM) and leukemic (ALL) cells [1,3].The Fold 2 of the three folds had the best ROC-AUC, indicating that some of the validation partitions might have more visually distinctive leukemic patterns [6,11]. This inter-fold variation is typical of hematological image analysis and is due to inherent heterogeneity in the dataset as opposed to model instability [6,11].**Figure 4** shows the ROC curves and AUC curves of the proposed supervised CNN Transformer architecture on the three validation folds, which shows the strength and the ability to generalize the model [22,23].

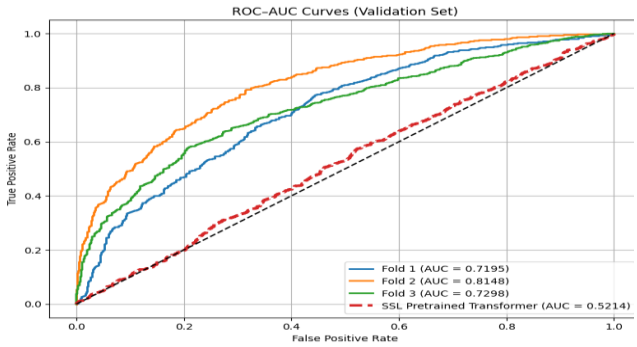


Fig. 2. *The hybrid CNN-Transformer has a significant advantage in performance compared to the SSL-pretrained transformer baseline, which produces the hybrid CNN-Transformer curve almost three stratifying folds (AUC: 0.71950,0.8148,0.7298).*

TABLE I. ROC-AUC Performance of the Hybrid CNN-Transformer Across Stratified Validation Folds

Fold	ROC–AUC
Fold 1	0.719
Fold 2	0.815
Fold 3	0.730
Mean ± SD	0.755 ± 0.043

Besides ROC-AUC, precision-recall (PR) was also conducted to additionally evaluate the performance of the model in case of class imbalance, which is prevalent in leukemia data [11,22,24]. PR -AUC focuses on how well positive class (ALL) detection can be relied upon as a joint measure of precision and recall [22,24]. The monitored CNN-Transformer model shows an overall high value of PR-AUC values in the validation folds, which shows that it is able to detect leukemic cells effectively while also keeping the number of false positives under control [24].

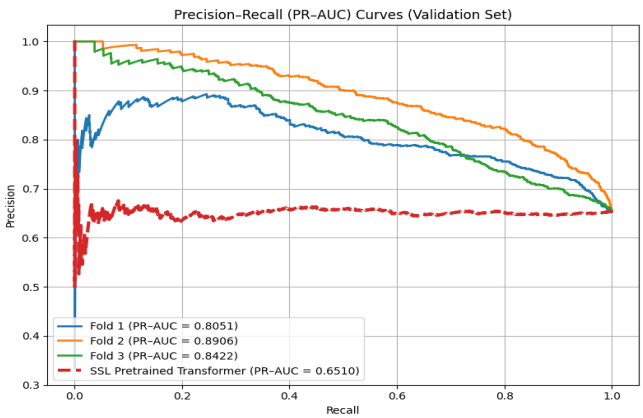


Fig 3. *The precision and better class separability of the hybrid CNN-Transformer is consistently superior to that of theSSL-pretrained transformer baseline as proved by PR curves over three stratified folds (PR-AUC = 0.8051-0.8906).*

TABLE II. PR-AUC Performance of the Hybrid CNN-Transformer Across Stratified Validation Folds

Fold	PR–AUC
Fold 1	0.8051
Fold 2	0.8906
Fold 3	0.8422
Mean ± SD	0.8460 ± 0.0351

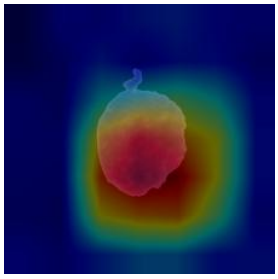
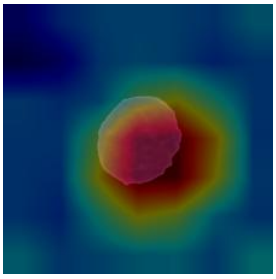
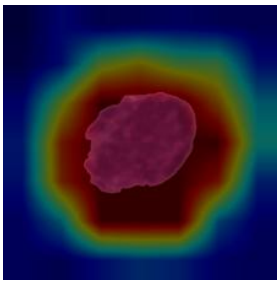
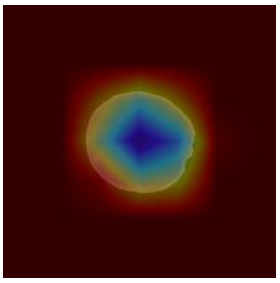
B. Comparison with Self-Supervised Pretrained Transformer :

In order to analyze the self-supervised representations in isolation, the self-supervised transformer without the downstream supervised fine-tuning was tested using the same validation criterion [18]. This model had a ROC-AUC of about 0.52 which is slightly higher than the random performance [22,23].This finding underscores a crucial feature of self-supervised learning models, that despite these models being useful in learning general-purpose visual representations, fine-tuning is needed to transfer these representations into clinically useful predictions [18,19]. The lack of marked supervision makes the SSL model unable to emulate leukemia-specific morphological findings that should be used to make a reliable diagnosis [1,3].Notably, the non-random nature of the performance of the unfine-tuned SSL model should not be understood as a drawback of the self-supervised learning itself [18]. Rather, it highlights the complementary nature of self-supervised pretraining, as well as, supervised optimization [18,19]. The comparison also confirms that the improvement in the performance of the proposed CNN-Transformer model is due to supervised learning that is task-oriented, and not necessarily the complexity of the architecture [18,19].

C. Qualitative Analysis and Explainability :

In addition to quantitative assessment, interpretability was also integrated to make it more clinically relevant [21]. The predictions of the model were considered by a combination of transformer attention maps and Grad-CAM visualization based on the CNN backbone [19,21]. As one can see in these visual explanations, the model is constantly sensitive to diagnostically relevant areas, such as nuclear boundaries, chromatin texture and cytoplasmic structure [1,3].In case of leukemic samples, patterns of activation are situated around swollen and distorted nuclei as opposed to the more diffuse and uniformly distributed attention of healthy cells [1,3]. These observations are in line with the established hematopathological criteria and qualitatively make evidence that the model has decision making process [1,3].The explainability mechanisms when combined provide a better understanding of the system and allow the domain experts to test predictions visually, which is essential in gaining trust and acceptance in clinical decision-support systems [21,1,3].

TABLE III. Grad-CAM and Transformer Attention Maps for Representative ALL and HEM Blood Smear Images

ALL	HEM
	
	

D. Result Explanation :

The visualizations of the grad-cam and transformer attention in Fig. 5 will give a qualitative understanding about the areas that affected the model predictions [21]. In the case of leukemic (ALL) samples, the model puts major focus on nuclear regions, but in the case of healthy (HEM) samples, the activation patterns are more compact and uniform in the activation attention maps [1,3]. This observation is agreeable with familiar morphological features employed in the hematological examination [1,3].The identified patterns in attention would indicate that the suggested CNN-Transformer model is based on the diagnostically significant areas of images but not on the background noise, which justifies the fact that the results of the classification can be interpreted [21].

E. Discussion :

The conducted experiment supports the claim that the developed hybrid CNN-Transformer model offers a powerful and explainable method of automated leukemia diagnosis on the basis of the peripheral blood smear images [18,21]. The supervised model is consistent in ROC-AUC performance between stratified validation folds, and therefore depends on variability and class imbalance of the dataset [11,22,23].The comparison to the self-supervised pretrained transformer demonstrates the need to use disease-specific classification problems to fine-tune with the supervision [18,19]. Although self-supervised pretraining provides a useful baseline of representation learning, its usefulness is only maximized by the use of labeled supervision, specific to the diagnostic goal [18].

A number of major observations can be made out of this study [18,21]:

I) **Generalization ability:** This indicates the possibility of successful learning of leukemia-relevant morphological features as shown by a consistent performance across folds [11,22,23].

II) **Clinical interpretability:** AttentionGracam fusion is a visual explanation that allows intuitive descriptions that are correlated with the diagnostic criteria of the experts [1,3,21].

III) **Methodological rigour:** Stratified cross-validation and realistic performance levels minimise the chances of optimistic bias or data leakage [22,23].

On the whole, the presented results suggest that hybrid CNN -Transformer models trained by supervised task criteria and exposed to explainability tools are a promising trend in medical image-based leukemia detection [18,21].

The further work will be aimed at supervised fine-tuning of self-supervised pretrained models, multi-class classification of leukemia subtypes, and testing on multi-center datasets to further test the robustness and clinical generalizability [18,19].

VI. Conclusion

This paper introduced an automated Acute Lymphoblastic Leukemia (ALL) detector based on CNN-Transformer on peripheral blood smear image [18]. The proposed algorithm can be successfully used to detect both the local or morphological features of the convolutional features and the greater or higher-order spatial relationships pertaining to the diagnosis of leukemia through a combination of convolutional feature extraction and transformer-based global context modelling [18,19]. Stratified three-fold cross-validation was used to test the model so as to have a fair and reliable test on realistic conditions of data distribution [22,23].The experimental findings indicate that the trained CNN–Transformer model can perform consistency in the discriminative performance of the model in the validation folds with ROC–AUC values significantly higher than random classification [22,23]. The inter-fold variation that is observed is an attribute of the heterogeneity of the hematological image information and not an attribute of the learning process being unstable [6,11]. On the contrary, the self-supervised pretrained transformer with no downstream fine-tuning performance is close to random, highlighting the need to use supervised optimization with respect to a clinical meaning of leukemia [18,19].In addition to the quantitative assessment, the qualitative assessment based on Grad-CAM and transformer attention overlays would offer insights into the decision making behavior of the model [21]. The visualizations show that the model mostly pays attention to diagnostically meaningful areas, especially nuclear morphology and chromatin structures, which justifies the interpretability and transparency of the proposed framework [1,3]. This explainability is essential to instill confidence in medical image analysis systems and the use of these systems in clinical processes [21].In general, the results imply that hybrid CNN-Transformer models (when trained with supervised tasks and developed with explainability models) can provide a promising and viable solution to the problem of

leukemia detection with microscopic image input [18,21]. Future directions will be on refined fine-tuning of self-supervised pretrained representations, multi-class leukemia subtype classification and testing in multi-center datasets to further consider robustness and generalizability in real-world clinical practices [18,19].

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